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PERSPECTIVE

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Conceptual and statistical problems with the DEC+J model of founder-event speciation and its comparison with DEC via model selection

Abstract

Phylogenetic studies of geographic range evolution are increasingly using statistical model selection methods to choose among variants of the dispersal-extinction-cladogenesis (DEC) model, especially between DEC and DEC+J, a variant that emphasizes "jump dispersal," or founder-event speciation, as a type of cladogenetic range inheritance scenario. Unfortunately, DEC+J is a poor model of founder-event speciation, and statistical comparisons of its likelihood with DEC are inappropriate. DEC and DEC+J share a conceptual flaw: cladogenetic events of range inheritance at ancestral nodes, unlike anagenetic events of dispersal and local extinction along branches, are not modelled as being probabilistic with respect to time. Ignoring this probability factor artificially inflates the contribution of cladogenetic events to the likelihood, and leads to underestimates of anagenetic, time-dependent range evolution. The flaw is exacerbated in DEC+J because not only is jump dispersal allowed, expanding the set of cladogenetic events, its probability relative to non-jump events is assigned a free parameter, *j*, that when maximized precludes the possibility of non-jump events at ancestral nodes. DEC+J thus parameterizes the *mode* of speciation, but like DEC, it does not parameterize the *rate* of speciation. This inconsistency has undesirable consequences, such as a greater tendency towards degenerate inferences in which the data are explained entirely by cladogenetic events (at which point branch lengths become irrelevant, with estimated anagenetic rates of 0). Inferences with DEC+J can in some cases depart dramatically from intuition, e.g. when highly unparsimonious numbers of jump dispersal events are required solely because *j* is maximized. Statistical comparison with DEC is inappropriate because a higher DEC+J likelihood does not reflect a more close approximation of the "true" model of range evolution, which surely must include time-dependent processes; instead, it is simply due to more weight being allocated (via *j*) to jump dispersal events whose time-dependent probabilities are ignored. In testing hypotheses about the geographic mode of speciation, jump dispersal can and should instead be modelled using existing frameworks for state-dependent lineage diversification in continuous time, taking appropriate cautions against Type I errors associated with such methods. For simple inference of ancestral ranges on a fixed phylogeny, a DEC-based model may be defensible if statistical model selection is not used to justify the choice, and it is understood that inferences about cladogenetic range inheritance lack any relation to time, normally a fundamental axis of evolutionary models.

1 | INTRODUCTION

In historical biogeographic studies that infer ancestral geographic ranges on phylogenetic trees, a particularly notable trend is the increasing use of statistical model selection, particularly in choosing between the dispersal-extinction-cladogenesis (DEC) model of range evolution (Ree, Moore, Webb, & Donoghue, 2005; Ree & Smith, 2008) and a variant called DEC+J (Matzke, 2014a). The impetus for DEC+J was a perceived shortcoming of DEC in island systems, namely, that it fails to consider "founder event" or "jump dispersal" speciation, in which new lineages are directly established by colonization without the intermediate existence of a widespread ancestor (Clark et al., 2008). DEC+J differs from DEC in allowing this kind of speciation scenario to be posited at internal nodes of the phylogeny, and includes a free parameter *j* that weights the contribution of such events to the likelihood of the data, observed species ranges at the tips of the tree.

Matzke (2014a) presented DEC+J as a more general case of DEC and argued that selection of DEC+J using statistical methods such as the likelihood ratio test or Akaike information criterion (AIC) constituted evidence in favour of jump dispersal as a biogeographic process. Numerous empirical studies have since followed this example and pitted DEC against DEC+J in testing hypotheses involving jump dispersal (e.g. Klaus et al. 2016; Fuchs, Pons, and Bowie, 2017; Su et al. 2016). However, these studies alone do not account for the trending use of model selection noted above. In fact, DEC+J is just one of several variants of DEC implemented in the "BioGeoBEARS" R package (Matzke, 2014b), and it is becoming increasingly common for studies to employ statistical methods of choosing among them without a biological reason for doing so—i.e. with no motivating hypothesis related to alternative processes of range evolution presented by the candidate models. A few examples (of many) are Cantalapiedra, Prado, Hernández Fernández, and Alberdi (2017), Brown et al. (2016), Givnish et al. (2016), Skeels and Cardillo (2017) and Kok et al. (2017). The application of model selection in these cases appears to be motivated by the same reasons one selects among models of nucleotide evolution for phylogeny reconstruction: that the best-fit model is necessary for accurate inferences.

This trend is unfortunate for two reasons: first, DEC+J is a poor model of founder event speciation, and second, unlike nucleotide models, DEC and DEC+J cannot be compared using standard

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statistical methods. Both of these assertions are consequences of a conceptual flaw common to both models, namely, that they include two distinct kinds of event to explain the data: anagenetic events of dispersal and local extinction that are probabilistic with respect to time, and cladogenetic events of range inheritance that are not. Cladogenetic events have greater explanatory power because unlike anagenetic events they can be invoked without having to account for their temporal probabilities of occurrence. However, in being dissociated from time they subvert the primary purpose of the evolutionary model and undermine its realism. DEC+J differs from DEC only by including jump dispersal as a kind of cladogenetic event. This exacerbates the problem of having non-equivalent kinds of event unfairly compete to explain the data, commonly to pathological levels where inferences converge exclusively on cladogenetic (jump dispersal) events. In such cases, rates of anagenetic dispersal and local extinction are estimated to be 0, and phylogenetic branch length (time) becomes irrelevant. Contrary to reasonable expectation, however, this does not guarantee that ancestral reconstructions are parsimonious with respect to the number of inferred events. The statistical non-equivalence of cladogenetic and anagenetic events means that DEC and DEC+J, which differ only in the relative allocation of explanatory power between event types, are inappropriate candidates for model selection. In the following sections, we explain these arguments in detail.

2 | BACKGROUND

2.1 | The Mk model: likelihood as a function of anagenetic change

DEC+J is derived from the DEC model, which in turn is derived from the standard Mk model of character evolution (Lewis, 2001). In the Mk model, discrete character states are assumed to change stochastically according to a process known as a continuous-time Markov chain (CTMC). The likelihood of the observed states of the terminal taxa having evolved on the given phylogeny is a function of evolutionary rates—the instantaneous probabilities of change from one state to another—and time, given by the branch lengths of the tree. It is computed using the well-known pruning algorithm of Felsenstein (1981), which recursively traverses the tree from tips to root and calculates, for each branch, the matrix of conditional probabilities for each ancestral state evolving to each descendant state. These probabilities of *anagenetic change*—stochastic transitions of a single lineage as it evolves through time—are explicitly defined by the model, and are obtained by integrating the constant transition rates over the duration of the branch.

At each internal node, the pruning algorithm computes a vector of *partial likelihoods*, which are the conditional likelihoods for each ancestral state given the node's descendant subtree and terminal states. These partial likelihoods vary because some ancestral states induce higher anagenetic probabilities, and thus explain the data better, than others. In general, states that require less anagenetic change (fewer transitions along branches) will have higher likelihood; indeed, the maximum-likelihood reconstruction of changes is often the most parsimonious reconstruction (Steel & Penny, 2000). Partial likelihoods are recursively combined to yield the global likelihood at the root node, integrating over all possible ancestral states at nodes, and sequences of change along branches of the tree.

2.2 | From Mk to DEC: Anagenetic and cladogenetic range evolution

The DEC model differs from Mk in three important ways (Ree & Smith, 2008). First, the states in DEC are not simple categorical values, e.g. {A,B} or {0,1}, as in Mk, but geographic ranges, coded as binary presence–absence values for a set of discrete areas. That is, a lineage can be endemic to a single area, or widespread across two or more areas (e.g. have states {A, B, AB}). Second, while Mk typically assumes that along a phylogenetic branch, any state can instantaneously change to any other state (e.g. A → B), in DEC, anagenetic range evolution is governed by two distinct processes: dispersal (range expansion, e.g. A → AB) and local extinction (range contraction, e.g. AB → B), whose time-constant rates d and e are free parameters. Instantaneous change is restricted to ranges that differ by a single dispersal or local extinction event (so transition from A to B requires two events, A → AB → B).

Third, and most important, is that while the Mk model assumes that all change is anagenetic—i.e. at internal nodes, the ancestral state is always inherited identically by the daughter lineages—the DEC model allows events of *cladogenetic range evolution*, in which widespread ancestral ranges are subdivided and inherited non-identically at nodes. More generally, the idea is that *range inheritance scenarios* occur when lineages divide, involving a triplet of states: the ancestral range prior to speciation, and the left and right daughter ranges derived from the speciation event. The scenarios for a minimal DEC model with two areas (A and B) and three possible ranges (A, B and AB) are shown in Figure 1a–c, grouped by the mode of range division: in situ (speciation of an endemic ancestor within a single area), vicariant (allopatric speciation of a widespread ancestor) and peripheral isolate (speciation of a widespread ancestor in a single area).

Range inheritance scenarios are equivalent to ancestral states in Mk. In the likelihood calculation for DEC (and its variants), partial likelihoods are computed for all distinct scenarios at each internal node. That is, at a given node, for every ancestral range, all of the scenarios for its subdivision and inheritance are evaluated, and each contributes to the likelihood through its induced probabilities of descendant anagenetic change.

The partial likelihoods are weighted to reflect the prior probability of range inheritance scenarios as stochastic outcomes, *given the occurrence of a speciation event*—the probability of the speciation event itself is not considered, a crucial point to which we will return. They are based on the assumption that at a node, distinct scenarios are not equiprobable a priori, but have a hierarchy of conditional expectations: for the ancestral range x_a, for the derived ranges $\{x_1, x_2\}$ produced by subdivision of the ancestral range, and for the sorting of those derived ranges into the daughter species $\{x_l, x_r\}$ the left

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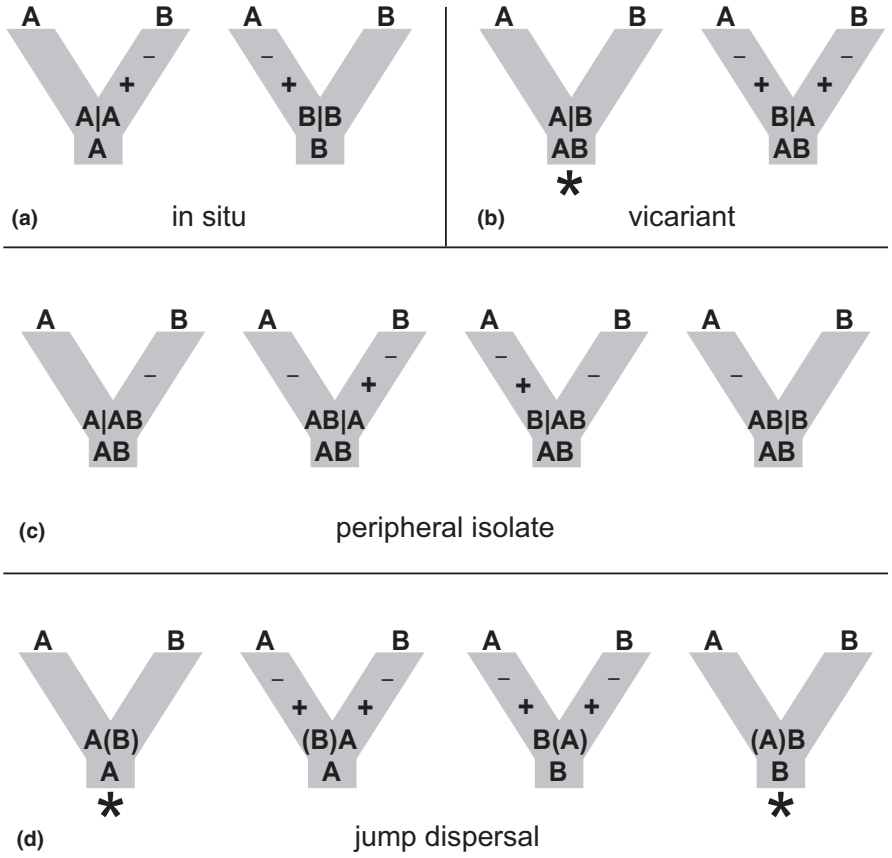


FIGURE 1 Range inheritance scenarios considered by DEC (a–c) and DEC+J (a–d) in relation to explaining sister species endemic to areas A and B, respectively. At the internal node (common ancestor), ancestral ranges are shown prior to (below) and following speciation (above). Subdivision of the ancestral range is indicated by |; jump dispersal is indicated by parentheses enclosing the area to which dispersal occurs. Along descendant branches, the minimum sequence of dispersal events (+) and local extinction events (−) required by the inherited ranges is shown. Scenarios that require no such events are highlighted by an asterisk.

and right branches descendant from the node. In general, the weight is

$$w(x_a \rightarrow x_l, x_r) = P(x_a)P(x_a \rightarrow \{x_1, x_2\})P(x_1 \rightarrow x_l, x_2 \rightarrow x_r)$$

In DEC, the weights are fixed, and based on flat priors at all levels of the hierarchy. Distinct ancestral ranges $x_a \in X$ are equiprobable; for a given range x_a, the distinct products of possible range subdivision events $\{(x_1, x_2)\}^{x_a}$ are equiprobable; and for a given pair of daughter ranges, assignment to left and right descendant branches is equiprobable (symmetric). Thus, if $k = |X|$ and $m_a = |\{(x_1, x_2)\}^{x_a}|$,

$$w(x_a \rightarrow x_l, x_r) = \frac{1}{2km_a}.$$

Continuing the two-area example, the weights for each scenario are given in Table 1.

2.3 | From DEC to DEC+J: cladogenetic range evolution by jump dispersal

Cladogenetic range evolution is the key to how DEC differs from Mk, and also to how DEC+J differs from DEC. In DEC, the only range inheritance scenarios allowed are those in which both

TABLE 1 Weights of range inheritance scenarios shown in Figure 1 for DEC and DEC+J. In DEC, the three possible ancestral ranges have equal prior probability, so the weights for each sum to $\frac{1}{3}$. In DEC+J, the weights of individual scenarios are functions of j, $0 \leq j \leq 3$. Priors on the ancestral range are thus variable; as $j \rightarrow 3$, $\Pr(\text{AB}) \rightarrow 0$ and $\Pr(\text{A}) = \Pr(\text{B}) \rightarrow \frac{1}{2}$

Ancestor	Cladogenetic event	Left	Right	DEC	DEC+J
A	In situ speciation	A	A	$\frac{1}{3}$	$\frac{1}{3} \cdot \frac{3-j}{3}$
A	Jump dispersal	A	B	–	$\frac{1}{12} \cdot j$
A	Jump dispersal	B	A	–	$\frac{1}{12} \cdot j$
B	In situ speciation	B	B	$\frac{1}{3}$	$\frac{1}{3} \cdot \frac{3-j}{3}$
B	Jump dispersal	B	A	–	$\frac{1}{12} \cdot j$
B	Jump dispersal	A	B	–	$\frac{1}{12} \cdot j$
AB	Vicariance	A	B	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$
AB	Vicariance	B	A	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$
AB	Peripheral isolate in A	A	AB	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$
AB	Peripheral isolate in A	AB	A	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$
AB	Peripheral isolate in B	B	AB	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$
AB	Peripheral isolate in B	AB	B	$\frac{1}{18}$	$\frac{1}{18} \cdot \frac{3-j}{3}$

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daughter ranges are *subsets* of the ancestral range (Figure 1a–c), reflecting the idea that speciation, modelled as an instantaneous event, is associated with subdivision, but not expansion or contraction, of the ancestral range. By contrast, DEC+J allows cladogenetic scenarios in which one daughter "jumps" to an area *outside* the ancestral range; these represent speciation events *caused by* dispersal to that area (Figure 1d). The model includes a free parameter j, the relative weight of jump dispersal scenarios in the likelihood calculation. The parameter is bounded such that weights are equal to those in DEC when $j = 0$, and place all weight on jump dispersal scenarios when $j = 3$ (Matzke, 2014a).

$$w(x_a \rightarrow x_l, x_r) = \begin{cases} \frac{1}{2km_a} \cdot j & \text{if} \{x_l, x_r\} \text{results from jump dispersal} \\ \frac{1}{2km_a} \cdot \frac{3-j}{3} & \text{otherwise.} \end{cases}$$

Weights for the two-area example are given in Table 1.

2.4 | Cladogenetic range evolution is not time dependent: a conceptual flaw, exacerbated in DEC+J

In DEC and DEC+J, two kinds of range evolution compete to explain the data, but they are modelled fundamentally differently. Anagenetic events of dispersal and local extinction occur as a CTMC, with probabilities that are explicit functions of rate and time (branch length). By contrast, cladogenetic events of range inheritance are not modelled as a CTMC, but are instead assumed to occur only at the internal nodes of the given phylogeny. They are probabilistic only in the sense of having stochastic outcomes *given a speciation event*; the speciation event itself has no time-dependent probability because internal nodes are treated as fixed, not as outcomes of a stochastic process of lineage diversification. We emphasize that neither model includes a parameter for the *rate* of speciation, even though they assign probabilities to distinct scenarios of range evolution at speciation events, and that in DEC+J, the j parameter is *not* a rate of founder-event speciation. Most importantly, by ignoring the time dependence of stochastic speciation events, the probabilities of range inheritance scenarios invoked at nodes are artificially inflated relative to anagenetic events. In the following sections, we explain why this is a conceptual problem for both models, but especially so for DEC+J, and why it argues against statistical comparisons of their likelihoods.

As statistical models of range evolution, DEC and DEC+J have a very specific purpose: to explain observed species ranges in terms of stochastic events, with event probabilities defined by parametric distributions that describe biogeographic processes. In a phylogenetic context where branch lengths measure the relative opportunity for change, it is natural (and well precedented by Mk) to model such events as occurring in a temporal dimension, i.e. as being stochastic with respect to continuous time. In this respect the dissociation of cladogenetic events from time represents a conceptual flaw in DEC and DEC+J (and other DEC variants): the extent to which data can be explained by events that are not probabilistic with respect to time represents a failure of the model to serve its basic purpose.

A problematic consequence of ignoring the time-dependent probabilities of cladogenetic events is that it allows for, and in fact favours, degenerate solutions: ancestral sequences at internal nodes that yield an unweighted conditional likelihood of 1 when rates of anagenetic range evolution are 0. In such cases, the data are explained perfectly, i.e. the model is essentially deterministic, with no component of statistical error, and completely divorced from a CTMC process. Consistent with time independence, the phylogenetic branch lengths then become irrelevant, as they are with parsimony. By contrast, likelihoods of solutions that require anagenetic events are never degenerate, and have theoretical maximum values that are generally orders of magnitude less than 1, simply due to the mathematics of integrating non-zero rates over time to obtain transition probabilities along branches. Cladogenetic events thus have a distinct advantage over anagenetic events in explaining the data, creating an ironic symptom of model failure: range inheritance scenarios that yield degenerate solutions also yield higher likelihoods.

2.5 | Degenerate likelihoods in DEC and DEC+J: A worked example

To illustrate degeneracy more concretely, consider the range inheritance scenarios shown in Figure 1 for our two-area example. The scenarios collectively represent the three possible ancestral ranges (A, B and AB) and all the patterns by which they yield left and right daughter ranges following speciation; those involving non-identical daughter ranges are naturally paired, reflecting the symmetry of bifurcation. Note that while most require at least one anagenetic event (dispersal and/or local extinction) along a descendant branch, one vicariance scenario and two jump dispersal scenarios require no anagenetic change at all: the inherited daughter ranges exactly match their extant descendant ranges (Figure 1b,d). For each of these scenarios, as rates of dispersal and local extinction approach 0, the unweighted partial likelihood (i.e. the likelihood conditional on the scenario at the node) approaches 1, while for all other scenarios the likelihood approaches 0. In other words, at the limit of no anagenetic range evolution, both DEC and DEC+J are degenerate: ancestral scenarios are either compatible with and perfectly explain the data, or the converse—they are incompatible with and entirely unable to explain the data.

The global likelihood, which considers all possible ancestral scenarios, is the weighted sum of partial likelihoods over the scenarios considered by the model (Table 1). For both models, the likelihood is maximized when anagenetic rates of range evolution are 0 (Figure 2). At this limit, the likelihood is simply the summed weights of the compatible scenarios. For DEC, this is a single vicariance scenario with a weight of 1/18; the log-likelihood is thus $\log\frac{1}{18} \approx -2.890$. For DEC+J, the likelihood is maximized when $j = 3$, which places all weight on the two compatible jump dispersal scenarios, each having weight $j/12 = 1/4$; the maximum log-likelihood is thus $\log(\frac{1}{4} + \frac{1}{4}) \approx -0.693$, an improvement over DEC by more than 2 log-units.

At this point, it is worth asking: should the mere observation of two sister species in different areas be interpreted as evidence in

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favour of ancestral jump dispersal over vicariance? It seems counter-intuitive, and yet that is the conclusion drawn by DEC+J.

In this example, both DEC and DEC+J are degenerate, and their maximum likelihoods simply reflect the weights assigned to compatible versus incompatible cladogenetic events. These results are invariant to time; whether the species diverged 1 or 1 million years ago is immaterial. To understand why cladogenetic events dominate the likelihood, consider a scenario that requires only one anagenetic event, e.g. the first scenario shown in Figure 1c. Its likelihood is the probability of A evolving to A along the left branch, multiplied by the probability of AB evolving to B along the right branch. For the latter to be positive an extinction event is required, and thus $e > 0$, but the effect of increasing e is also that both branches (especially the left) are less likely to survive to the present, illustrating how anagenetic processes are time-dependent. For branches of length 1, the maximum unweighted likelihood for this scenario is log (0.148) = −1.944 for $\hat{e} = 0.405$. This is almost an order of magnitude lower than for scenarios requiring no anagenetic events, which have maximum unweighted (conditional) likelihoods of 1.

2.6 | Pathological consequences of degeneracy in DEC+J

DEC+J differs from DEC in expanding the scope and influence of cladogenetic range evolution, both by allowing jump dispersal scenarios and assigning a free parameter to their partial likelihood weights. For this reason it is generally far more prone to degeneracy than DEC, favouring inferences dominated by cladogenetic jump dispersal scenarios and rates of anagenetic change that approach or equal 0; in these cases the DEC+J likelihood can exceed that of DEC by many orders of magnitude. This pattern was noted in the original paper introducing the model (Matzke, 2014a), and is prevalent in subsequent literature (e.g. Cantalapiedra et al., 2017; Klaus, Morley, Plath, Zhang, & Li, 2016; Norhazrina, Vanderpoorten, Hedenäs, & Patiño, 2016).

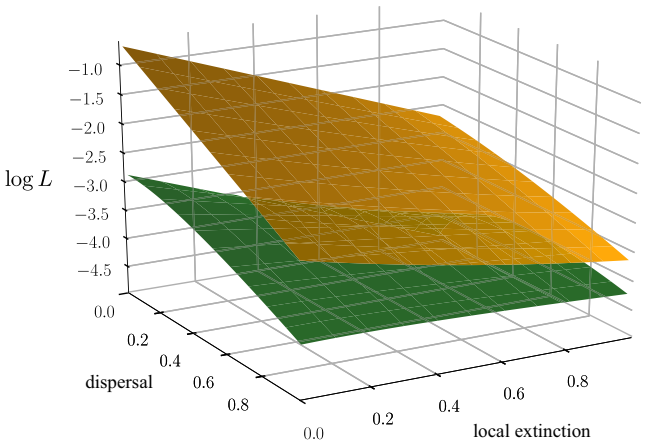


FIGURE 2 Likelihood surfaces for DEC (lower) and DEC+J (upper), for sister species endemic to areas A and B (Figure 1). The DEC+J surface is plotted for $j = 3$ (only jump dispersal scenarios are considered) because the likelihood is maximized at this value. Both models have maximum likelihood when rates of dispersal and local extinction are 0, at which point they are also degenerate: the likelihood is fully determined by the weights of the ancestral range inheritance scenarios that are compatible with the data; all other scenarios have zero likelihood. For DEC, there is one compatible scenario, with weight 1/18. For DEC+J, two jump dispersal scenarios are compatible, each with weight 1/4 (Table 1). The difference in likelihood is thus due only to the different weights assigned to range inheritance scenarios at the ancestral node by each model, and not to one model providing a finer-grained description of a stochastic process, as assumed by model selection methods such as AIC. [Colour figure can be viewed at wileyonlinelibrary.com]

While it is tempting to view such inferences as "parsimonious," in the sense that parsimony is not an explicitly probabilistic model and is invariant to time (e.g. Maddison, 1995; Schluter, Price, Mooers, & Ludwig, 1997), DEC+J inferences can in fact be highly unparsimonious with respect to inferred sequences of range evolution. Consider a slightly expanded version of the disjunct sister species example, with two additional species endemic to B added in a pectinate topology (Figure 3). Range inheritance scenarios at three internal nodes now require evaluation. In DEC, there is one ancestral sequence of scenarios that requires no anaegnetic change (Figure 3a), while in DEC+J there are four others (Figure 3b–e), two of which involve jump dispersal events at every node (Figure 3b,c). Maximizing the DEC likelihood yields log $L = -8.50$ at $d = 0.060$, $e = 0$. The non-zero rate of dispersal means that inference does not converge exclusively on the sequence in Figure 3a; sequences involving anagenetic dispersal also contribute to the optimal likelihood. In fact, setting $d = e = 0$ yields log $L = -8.67$; $\Delta\log L = 0.17$. For this example we can thus see that DEC does not completely reject anagenetic (continuous-time) processes.

By contrast, with DEC+J the likelihood is maximized (log $L = -3.47$) at $j = 3$, $d = e = 0$. As with the sister-species example, inferring both anagenetic rates to be 0 indicates that the likelihood does not reflect any time-dependent process. Moreover, that j is maximized indicates that ancestral sequences involving any non-jump scenario, e.g. endemic speciation in area B, are removed from consideration, even those requiring no anagenetic change (Figure 3d,e). That is, inference converges on the two sequences involving jump dispersal at every node (Figure 3b,c), which are maximally unparsimonious from the standpoint of ancestor-descendant changes. By contrast, the most parsimonious sequence invokes only one jump dispersal event at the most recent internal node (Figure 3e). DEC+J yields a likelihood 5.04 log-units greater than DEC, and is the model favoured by AIC (uncorrected; $\Delta AIC = 9.08$). However, it also yields inferences that reject a time-dependent evolutionary process, and converges on ancestral reconstructions that are in direct opposition to parsimony and, we would argue, intuition and common sense.

If degeneracy is taken as a general indicator of potentially unreasonable inferences, then DEC+J should be viewed with acute concern. Degenerate solutions are possible in a worryingly large set of cases, including (but not limited to) all those in which all species are single-area endemics and no sister species are endemic to the same area—the latter because that case cannot be explained by jump dispersal. The potential for degeneracy with DEC+J is generally predictable, increasing with the propensity for species to be single-area

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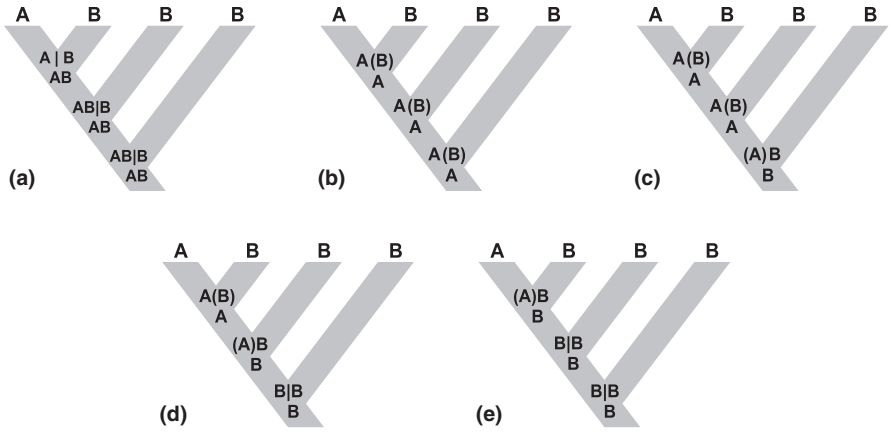


FIGURE 3 Ancestral sequences of range inheritance requiring no anagenetic change under DEC (a) and DEC+J (a–e). In sequences b and c, jump dispersal occurs at every internal node, while sequences d and e involve one and two in situ speciation events respectively. The DEC+J likelihood is maximized (log $L = -3.47$) at $j = 3$, $d = e = 0$, indicating convergence on sequences b and c, in which jump dispersal occurs at every ancestral node. Sequences d and e, which invoke fewer jump dispersal events and in that sense are more parsimonious, are excluded from contributing to the likelihood because they include speciation events within B, which have zero probability when $j = 3$. By contrast, the DEC likelihood is maximized (log $L = -8.50$) at $d \approx 0.060$, $e = 0$, indicating that some weight is given to sequences involving a root scenario of B| B and a single range expansion event along an internal branch (not shown)

endemics, and with the probability that sister species differ in their ranges. This is because the establishment of any single-area endemic is easily explained by jump dispersal from an ancestor not inhabiting that area. However, it is worth considering how other processes, unrelated to jump dispersal, could generate similar patterns of data prone to degenerate solutions with DEC+J.

Degeneracy is a symptom of anagenetic and cladogenetic events competing to explain the data, with cladogenetic events having an unfair advantage because they lack time-dependent probabilities. This explains why anagenetic parameters are chronically underestimated in simulations (Matzke, 2014a; Ree & Smith, 2008). In terms of likelihood, cladogenetic events are less costly to invoke (see below), and are therefore preferred in optimizations; this has the effect of hiding anagenetic events that actually occurred, lowering their rate estimates.

2.7 | Likelihoods from DEC and DEC+J are not statistically comparable

Thus far we have argued that DEC and DEC+J are both flawed because they model cladogenetic range evolution but fail to account for the process (time-dependent probabilities) of cladogenesis itself; and that DEC+J exacerbates this flaw to the extent that we question its usefulness for meaningful inference. Here, we explain how this relates to model selection—specifically, how the difference between DEC and DEC+J is distinctly unlike the difference that normally characterizes candidate models in applications of model selection based on AIC, and how this in turn casts serious doubt on whether it is appropriate to compare their likelihoods by this criterion.

If we consider the most common applications of model selection in phylogenetics, whether by AIC or other means, it is clear that in all cases, the candidate models offer alternative parameterizations of a stochastic process *in which the generated events occur along a temporal axis*. In this context, time represents a relative measure of the opportunity for change. For example, all models of DNA sequence evolution describe the probability of nucleotide substitution as a function of time; all models of lineage diversification describe the probabilities of speciation and extinction as a functions of time. There is thus a fundamental equivalence in how competing models explain the data: the basic units of currency, probabilities of evolutionary events, are all expressed in relation to time.

In DEC and DEC+J, anagenetic and cladogenetic events of range evolution are decidedly not equivalent in this sense: as already shown, the former are probabilistic with respect to time, and the latter are not. This critical difference can be thought of in terms of a cost analogy, in which the goal is to explain the data as cheaply as possible (i.e. with highest likelihood). Anagenetic events are expensive because they "cost" (are a product of) two probability factors: the probability of change as a function of time, and the probability of the outcome of that change. By contrast, cladogenetic events are cheap, costing only 1 probability factor—the outcome of range inheritance at a fixed internal node—because the time-dependent probability is ignored.

DEC+J differs from DEC only in its greater capacity to explain the data via "cheaper" cladogenetic events. This is fundamentally different from every other case of model selection in phylogenetics, in which the competing models differ only in how their parameters modify the probabilities of time-dependent evolutionary events. The DEC/DEC+J case is thus unique, and we are not aware of any precedent that sheds light on whether the application of conventional model selection methods is appropriate. However, there are compelling signs to the contrary.

The fact that higher likelihoods in DEC+J result from ignoring time-dependent processes, and not from improving the parameterization of those processes, is the most prominent red flag. Time is a universal axis across parametric evolutionary models because it is

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fundamental to evolution itself. It is difficult (we would say impossible) to imagine a probabilistic model of evolution that does *not* incorporate time in a meaningful way; so encountering one that increases its likelihood by dissociating its process from time should raise immediate concerns about its connection to reality.

More specifically, AIC measures the relative degree to which a candidate model *loses information* in approximating the true (unknown) generating model (Burnham & Anderson, 1998). It should be uncontroversial to assert that any "true" model of geographic range evolution with phylogenetic branching should certainly include, if not depend fundamentally on, time-dependent processes. Indeed, there is abundant empirical evidence indicating that rates of dispersal and extinction are non-zero and variable through time (Silvestro et al., 2016). But when DEC+J "wins" the AIC contest, it does so by minimizing such processes and effectively losing information about time. That an evolutionary model can be judged closer to reality by dissociating itself from time strongly suggests that the measure of distance (AIC) is not functioning as intended. In this case, we know why: the DEC+J likelihood is artificially inflated relative to DEC.

A potential counterargument to this view is that AIC selection of DEC+J simply indicates that the observed branch lengths on the phylogeny fit the data poorly—that evolutionary change is simply concentrated at cladogenetic events, similar to testing punctuational hypotheses by transforming branch lengths with κ (Pagel, 1999). However, this analogy fails because DEC+J is a probabilistic model of speciation, which calls for considering the observed phylogeny as the outcome of a stochastic speciation process. DEC+J does not do this; it parameterizes only the mode of speciation, not its rate. It does not improve the fit of data to a phylogeny so much as ignore fundamental probabilities in the process it is trying to model.

2.8 | DEC+J is not, in fact, a fully probabilistic model (but could be)

Up to this point, we have described cladogenetic range evolution in DEC and DEC+J as probabilistic (although not with respect to time) because range inheritance scenarios are considered to be stochastic outcomes of speciation events. However, it is worth noting that this is not strictly true for DEC+J as described by Matzke (2014a). To be fully probabilistic, the outcomes at a speciation event must have probabilities (weights) that sum to 1, or a constant (e.g. $1/k$), *for every ancestral range*. In DEC+J, weights of scenarios that do not involve jump dispersal are multiplied by $(3-j)/3$, which approaches 0 as $j \rightarrow 3$. This incorrectly penalizes the widespread range for which no jump dispersal scenarios are possible. In our two-area example, the summed weights of the ancestral range AB are less than 1 if $j > 0$, and approach 0 as $j \rightarrow 3$ (see Table 1).

The solution to this problem is to define j explicitly as the relative probability of a jump dispersal scenario versus a non-jump scenario as the outcome of a speciation event, i.e. $0 \leq j \leq 1$, and normalize the weight of any given range inheritance scenario by a term z that reflects the fraction of jump dispersal scenarios possible for the ancestral range. The weights for jump dispersal and non-jump scenarios would thus be scaled by j/z and $(1-j)/z$, respectively, where $z = mj+n(1-j)$ and m and n are the number of jump and non-jump scenarios possible, with the weight equal to 1 if $m = 0$ and $j = 1$. This logic is currently being implemented in the software package RevBayes (Höhna et al., 2016), allowing j, or indeed parameters for any set of cladogenetic event categories, to be defined as a simplex (M. Landis, personal communication).

Redefining j in this manner does not mitigate our view that DEC+J is a poor model of jump dispersal, or compromise the argument that model selection with DEC and DEC+J is statistically inappropriate. The problem remains that cladogenetic events lack time-dependent probabilities and thus have unjustifiably greater explanatory power than anagenetic events.

2.9 | How to model jump dispersal in continuous time: ClaSSE

Thus far we have argued that DEC+J is not a particularly reasonable way to model jump dispersal, and that comparing its likelihood to DEC is not statistically justified. What then is the alternative?

If the question is how to model phylogenetic range evolution with jump dispersal as a stochastic process, then the answer is ClaSSE (Goldberg and Igić, 2012), a generalized state-dependent speciation-extinction model in which probabilities of anagenesis and cladogenesis are jointly considered as part of a CTMC. (The GeoSSE model of range-dependent diversification (Goldberg, Lancaster, & Ree, 2011) is a special case of ClaSSE.) Applying ClaSSE to range evolution requires defining the states (geographic ranges) and specifying the transition rates for both anagenesis and cladogenesis. The former would follow DEC in allowing range expansion and contraction by dispersal and local extinction. The latter would model range-dependent speciation: for each range, one could assign a free rate parameter to every possible mode of range inheritance, including jump dispersal. The final consideration is global lineage extinction: to follow DEC, this rate should be non-zero only for single-area ranges.

ClaSSE was criticized by Matzke (2014a) as being too parameter-rich and therefore likely to lack statistical power. While adequate sample sizes are certainly needed (e.g. Davis, Midford, & Maddison, 2013), it is important to appreciate that while the theoretical maximum number of rate parameters does indeed increase geometrically with the number of areas, the number used in practice may be much less. A minimal model in which diversification is independent of geographic range could have only three rates, for dispersal, local extinction and speciation. To test hypotheses about jump dispersal specifically, one additional parameter could be added, scaling the rate at which it occurs relative to the background speciation rate. Such models could be compared using conventional criteria such as AIC. However, caution is warranted: simulations have shown that inferences from state-dependent diversification models can have elevated Type I error rates because they assume that all heterogeneity in diversification rates is due to the modelled process (Rabosky & Goldberg, 2015); Beaulieu and O'Meara (2016) offer a potentially mitigating solution, at the cost of greater model complexity.

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Needless to say, ClaSSE likelihoods cannot be compared with DEC and DEC+J. In ClaSSE, the phylogeny is a random variable with a probability distribution that accounts for "ghost" lineages—extinct and hence unobserved branches. For a given data set, the ClaSSE likelihood will thus always be lower than DEC and DEC+J, which treat the phylogeny as fixed, and whose likelihoods only reflect probabilities of lineage survival between internodes.

The main impediment to using ClaSSE for historical biogeography seems to be mainly one of convenience: the model is currently implemented only in the "diversitree" R package (FitzJohn, 2012), with no functions to help automate the conversion of anagenetic and cladogenetic rates of geographic range evolution into the general form required by the software. These are minor obstacles that if removed would facilitate greater exploration of the utility of ClaSSE-based models in testing hypotheses of range-dependent diversification.

2.10 | Recommendations for using DEC-based models in practice

All models are incomplete descriptions of reality and thus "wrong" in some sense (Box, 1976). In the case of DEC and its variants, treating the phylogeny as fixed and failing to model cladogenetic range evolution as a stochastic process in continuous time is a conspicuous deficiency in realism that, as we have seen, can have undesirable consequences such as degeneracy and counterintuitive deviations from parsimony. It is certainly the case that statistical model selection between DEC and DEC+J should not be done if one wishes to "test for" jump dispersal as a meaningful process of range evolution. In general, hypotheses involving the geographic mode of speciation require models in which speciation, and thus the phylogeny itself, is treated as the outcome of a stochastic process in continuous time, i.e. range-dependent diversification (Goldberg et al., 2011; Ree & Sanmartín, 2009). One can even imagine co-estimating the phylogeny itself with a model that includes such a process (e.g. Landis, 2017).

However, often the primary goal is simply to infer ancestral ranges on a given tree. In such cases, speciation mode and diversification dynamics are not central to any biogeographic hypothesis of interest, and are thus effectively nuisance parameters. A DEC-based model that does not parameterize the mode of speciation (i.e. does not include *j* or a similar parameter) may be a defensible choice here, so long as it is not selected using statistical methods such as AIC that assume probabilistic equivalency of events, as all DEC-based models differ only in their allocation of explanatory power between non-equivalent event classes. Unfortunately, in our editorial experience with the *Journal of Biogeography*, model selection between DEC and DEC+J (and other DEC variants) has become to be perceived as almost a de facto requirement, either requested by reviewers, or done seemingly to preempt such requests. Our purpose here is to show why this is neither necessary nor recommended. More generally, we also wish to encourage greater critical attention to the structure and assumptions of models used in parametric historical biogeography.

In using DEC-based models, concerns about which particular model best fits the data should be tempered by the realization that a single trait (geographic range) at the tips of a tree offers relatively little information about its generating process (e.g. Cunningham, Omland, & Oakley, 1998). If different models confidently yield conflicting reconstructions that are meaningful to the study, it seems entirely reasonable to favour one over another by making arguments based on empirical (biological, geographic) considerations—in other words, to allow non-statistical judgements guide model choice. Most importantly, researchers should understand and appreciate that DEC-based models yield inferences about cladogenetic range evolution that lack any relation to time, normally a fundamental axis of evolutionary models, and consequently exercise appropriate caution in drawing conclusions about ancestral range dynamics.

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